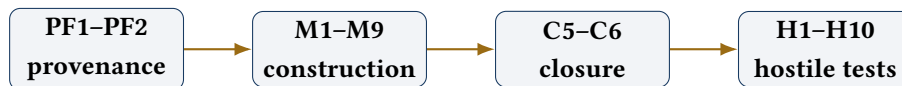


Solvency Field Theory

Bell Completion Derivation Canon

From Local Pair Formation to H1–H10 Hostile Closure

Audited internal source-of-truth edition



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5.6 SOL – derivation, formalization, and audit

Internal derivation canon, version 1.0

July 2026

This document records the conditional mathematical chain by which the SFT Bell program moves from a selected local zero-total-frame source sector to the ideal singlet probability law and then through ten adversarial closure gates. It is the technical source of truth for a separate public narrative. It is not a claim that the microscopic source sector has already been derived from first principles, and it is not a claim of experiment-specific loophole closure.

Abstract

The SFT Bell program asks a deliberately narrow question: can a framework built from locally real particle supports, irreversible history writes, local detector contexts, and no preferred spacelike measurement order reproduce the singlet correlation law without inserting the angular kernel, writing a hidden counterfactual answer table, admitting superluminal signaling, or silently using measurement dependence or postselection?

The construction is conditional on a prepared pure $L = 0, S = 0$ two-support source sector. The associated reduced two-state zero-total-frame projector is

$$\Pi_0 = \frac{1}{4} (\mathbb{I} - \sigma_A \cdot \sigma_B). \quad (1)$$

Local rotational covariance, axis reversal, and sharp binary closure then force

$$D(\mathbf{n}) = \mathbf{n} \cdot \boldsymbol{\sigma}, \quad P_s(\mathbf{n}) = \frac{1}{2} (\mathbb{I} + s \mathbf{n} \cdot \boldsymbol{\sigma}). \quad (2)$$

For one actually selected context $(\mathbf{a}, \mathbf{b})$, direct composition gives

$$\mu_{st}(\mathbf{a}, \mathbf{b}) = \text{Tr}[\Pi_0(P_s(\mathbf{a}) \otimes P_t(\mathbf{b}))] = \frac{1}{4} [1 - st \mathbf{a} \cdot \mathbf{b}]. \quad (3)$$

Positive orthogonal additivity forces relative branch intensity to be linear in this support measure. Ideal normalized completion therefore yields

$$\mathbb{P}(s, t \mid \mathbf{a}, \mathbf{b}) = \frac{1}{4} [1 - st \mathbf{a} \cdot \mathbf{b}], \quad E(\mathbf{a}, \mathbf{b}) = -\mathbf{a} \cdot \mathbf{b}. \quad (4)$$

Local branch-record maps commute on disjoint spacelike factors, so A-first and B-first descriptions produce one invariant common-future ledger join. The ideal distribution is operationally no-signaling. Measurement independence remains an explicit Bell-test premise, not a consequence of notation or a theorem of finite data.

The frozen distribution is then attacked. It lies outside the measurement-independent Bell-local polytope, reaches $2\sqrt{2}$ for a fixed CHSH witness, survives arbitrary normalized local null-complete detector processing, quantifies the measurement-dependence escape, remains frame-order and delayed-choice consistent, supports entanglement swapping through a later local middle interaction, requires genuine N -party rela-

tional support for GHZ source sectors, yields exact noisy-family threshold surfaces, and exposes the final detection-loophole boundary.

Audited final statement

SFT preserves causal locality and operational no-signaling, but under measurement independence its frozen ideal Bell distribution is not Bell-local. The ideal and named noisy record architectures survive H1–H10 without superluminal signaling, preferred foliation, negative probabilities, hidden source-setting dependence, or unacknowledged postselection. The preparation of the selected source sectors, the microscopic intensity and amplification dynamics, environmental rates, and experiment-specific loophole closure remain explicit open debts.

Canonical proof-status ledger

Load-bearing step	Status	Exact boundary
Pure $L = 0$, $S = 0$ source preparation	Premise / source debt	The Bell derivation conditions on this selected channel; it does not derive its branching fraction or microscopic production operator.
Zero-total-frame projector Π_0	Conditional theorem	Exact representation of the selected two-state $S_{\text{tot}} = 0$ sector.
Local analyzer $D(\mathbf{n})$	Conditional theorem	Unique for a sharp, calibrated, rotationally covariant two-state analyzer.
Joint support law μ_{st}	Theorem	Direct trace consequence of Π_0 and the local projectors.
Relative intensity $I_{st} = \kappa \mu_{st}$	Conditional theorem	Follows from nonnegative orthogonal additivity; the absolute rate and material dynamics remain open.
Ideal sign law $P_{st} = \mu_{st}$	Conditional theorem	Requires ideal completed trials or an explicit null-complete extension.
Null-complete no-signaling	Theorem	Preserved by every normalized local response channel when all records are retained.
Measurement independence	Premise / audit target	$P(a, b \mid \Lambda_{\text{src}}) = P(a, b)$ is not derived from schema separation.
Bell-local impossibility	Theorem under the premise	The frozen distribution has no nonnegative measurement-independent local factorization.
Laboratory loophole closure	Data/protocol debt	Requires a declared trial ensemble, nulls, singles, timing, background, stopping, and setting audits.

Version 1.0 audit note

Version 1.0 preserves every frozen gate result and hash. It tightens the document-level mathematics and claim boundaries. The principal changes are:

1. the legal pair object is defined as an intersection subspace; a noncommuting projector sandwich is no longer called a projector;
2. the isolated one-null-support source claim is restricted to cases with no recoil or external support;
3. the local record branch maps are written on the correct tensor factors;
4. the intensity-linearity argument is upgraded to a complete monotonic-additive proof;
5. unsharp POVM effects are separated from the orthogonal sharp-branch theorem;
6. the H3 selected marginal quantity is identified correctly as a signed-expectation shift;
7. the H4 mutual-information optimum receives an explicit symmetry-reduction proof sketch;
8. the H7 middle projector is embedded without tensor-order ambiguity;
9. the noisy GHZ Mermin formulas use the correct absolute-value boundary;
10. the final theorem is split into ideal-kernel, Bell-exclusion, and empirical-inference layers.

This canon uses native SFT language first. Standard quantum-information notation is the exact representation used to prove the reduced support claims; it is not offered as a replacement ontology. The probability interpretation is introduced only after the separate M8 intensity bridge.

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Frozen rules, provenance, and proof obligations

1.1 Non-negotiable construction rules

The Bell program was constructed under the following frozen restrictions:

```
NO_QUANTUM_KERNEL_IMPORT,  
NO_MINUS_COSINE_INSERTION,  
NO_TARGET_DERIVED_TUNING,  
NO_MEASUREMENT_DEPENDENCE_UNLESS_EXPLICITLY_ADMITTED,  
NO_PREFERRED_FIRST_MEASUREMENT,  
NO_SUPERLUMINAL_SIGNAL,  
NO_PRIVATE_HISTORY_AS_EXCHANGE_OR_OUTCOME_LABEL,  
NATIVE_SFT_FIRST,  
BELL_QM_TRANSLATION_SECOND.
```

The rules matter because the target law is familiar. A derivation that inserts $-a \cdot b$, assumes the completed-event probability rule at source level, or imports a pre-existing answer table would only redescribe the target.

1.2 Master write architecture

The working operator order is

$$\Pi_{\text{allowed}} U_{\text{support}}[\Gamma \cdot \Pi] \Pi_{\text{allowed}} \longrightarrow U_{\text{branch}} \longrightarrow U_{\text{history/settlement}} \longrightarrow \text{readouts}. \quad (1.1)$$

The legality projector and support evolution need not commute:

$$[\Pi_{\text{allowed}}, U_{\text{support}}] \neq 0. \quad (1.2)$$

This noncommutation records boundary rejection and support restriction. It is not treated as a universal substitute for all dynamics.

1.3 Ontology and causal boundary

The construction uses only the existing SFT stack:

- locally real particle supports with irreversible no-overwrite histories;
- closure frames and branch orientations;
- local source vertices and retarded support propagation;
- local detector apparatus and record registers;
- shared relational support as a restriction on jointly legal histories;
- common-future ledger comparison.

No new mediator, field, dimension, global collapse hypersurface, or preferred foliation is added.

1.4 Representation boundary

The reduced Hilbert-space and Pauli-matrix representation is used because the selected closure support is two-state and because the source restriction is a one-dimensional zero-total-frame sector. Within that representation, projector identities and traces are exact.

The representation does *not* by itself supply a probability interpretation. The derivation deliberately separates:

$$\text{source projector} \longrightarrow \text{support overlap} \longrightarrow \text{relative intensity} \longrightarrow \text{completed-event probability}. \quad (1.3)$$

The trace at M6 is first interpreted as squared legal support. Its probability status is earned only after the M8 additivity and completion premises.

1.5 Provenance audit

The pull-forward and provenance gates established three decisive facts.

First, the earlier V13.5 singlet section contained a real conditional theorem: rotational invariance, unbiased local restrictions, perfect same-axis anticorrelation, contextual local readout, and a bilinear representation imply the singlet law. However, that text also displayed the target angular intensity before the microscopic support and intensity bridge were fully earned. The Bell program therefore could not cite the old target equation as an input.

Second, the write-race normalization machinery can normalize supplied nonnegative intensities:

$$P(i) = \frac{I_i}{\sum_j I_j}, \quad (1.4)$$

but it does not determine the angular shape of I_i . The shape had to be derived from the source and local analyzer geometry.

Third, the old line

$$I_{st} \propto 1 - st \mathbf{a} \cdot \mathbf{b} \quad (1.5)$$

was quarantined as a target insertion until M6–M8 independently reproduced it.

PF1–PF2 provenance result

The pair projector and local analyzer route was authorized only if the source object, local projectors, joint support measure, and native intensity bridge were independently derived. The old angular kernel could be used only as a regression target after the new chain had closed.

1.6 Types of statement

Throughout the canon:

- **Theorem** means derived from the frozen mathematical premises of the gate.
- **Conditional theorem** means exact given a named source channel, detector class, or completion premise that remains physically open.
- **Premise** means explicitly assumed and not derived, such as measurement independence.
- **Hostile kill** means an alternative model or interpretation was constructed and shown inconsistent with the frozen requirements.
- **Open debt** means a missing microscopic mechanism, calibration, source model, or experiment-specific audit.

1.7 Dependency map

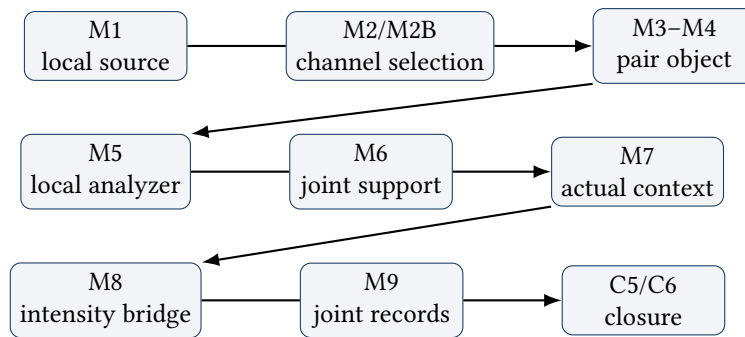


Figure 1.1: Construction dependency order. Each stage freezes before the next is scored.

Local pair formation and source-channel selection

2.1 M1: local pair-source reconstruction

A candidate pair source must be reconstructible from one local formation event with outgoing supports that propagate causally. Let k_A and k_B be two future-directed null support momenta before massive closure. Then

$$P^\mu = k_A^\mu + k_B^\mu, \quad P^2 = 2k_A \cdot k_B. \quad (2.1)$$

For nonparallel future-directed null supports,

$$P^2 > 0, \quad (2.2)$$

so the two-support system possesses a timelike center-of-momentum invariant.

An *isolated* one-null-support route cannot by itself supply that invariant, and a parallel two-null-support route also has $P^2 = 0$. This statement does not exclude physical pair-production processes in which an external field, nucleus, recoil body, or additional support participates; those extra degrees of freedom are part of the local source event and its conserved momentum ledger. A relation posited before any local contact fails the local-source requirement.

The surviving M1 source geometry is therefore:

1. one local common event v , including any required recoil or external support;
2. two outgoing causal particle supports;
3. a nonzero timelike invariant for the complete local source system;
4. a permanent common ancestry record H_v ;
5. opposite charge/framing winding balance where the selected production channel requires it.

M1 source theorem

A massive two-support pair is not supplied by an isolated single null support, by parallel null supports, or by a relation written before local contact. A complete local source event with the required conserved support and nonparallel outgoing branches supplies the minimal causal ancestry object.

2.2 M2: what no-overwrite does and does not force

A literal braid was initially considered as the common-history mechanism. No-overwrite gives a one-way result:

$$\text{if a nontrivial braid class is written, it cannot later be erased.} \quad (2.3)$$

It does not give the converse:

$$\text{individual } 4\pi \text{ closure does not force a nonzero mutual braid.} \quad (2.4)$$

A separable post-contact countermodel survives: two individually legal 4π supports can leave the same source vertex without nonzero centerline winding. Therefore generic pair production does not earn mandatory literal braid topology.

2.3 M2B: selected channel rather than generic production

The source restriction is channel-specific. For two closure supports, define

$$\mathbf{J}_{\text{tot}} = \mathbf{L}_{\text{rel}} + \mathbf{S}_A + \mathbf{S}_B. \quad (2.5)$$

The pure prepared channel used by the Bell construction is

$$L = 0, \quad S_{\text{tot}} = 0, \quad J_{\text{tot}} = 0. \quad (2.6)$$

Neither head-on equal energy nor $J_z = 0$ alone implies this channel. Even $J = 0$ need not imply spin singlet when orbital angular momentum is available. The source selector must explicitly identify the pure $L = 0, S = 0$ branch.

For two spin- $\frac{1}{2}$ supports, the $S_{\text{tot}} = 0$ subspace is one-dimensional and contains no product state. The selected channel is therefore intrinsically relational.

Critical source premise

The Bell derivation does not derive that an arbitrary pair source enters this channel. It conditions on a source preparation whose selected branch is pure $L = 0, S = 0$. The branching fraction, preparation fidelity, mode purity, and microscopic source operator remain external source-generation debts.

2.4 Relative pair-frame holonomy

The surviving native replacement for mandatory centerline braiding is a relative pair-frame datum. Let F_A and F_B be retarded closure-frame transports from the source. Their relative class is

$$R_{AB} = F_A^{-1} F_B. \quad (2.7)$$

Under a common source-frame relabeling $F_A \mapsto GF_A, F_B \mapsto GF_B$,

$$R_{AB} \mapsto (GF_A)^{-1}(GF_B) = R_{AB}. \quad (2.8)$$

Thus the pair may carry a shared relational frame object without a literal spatial cable or knot between the particles.

2.5 Source data frozen for the construction

The M3 input is the tuple

$$\mathcal{S}_{AB} = (H_v, \chi_{L=0, S=0}, F_A, F_B, [R_{AB}], \Gamma_A, \Gamma_B), \quad (2.9)$$

where Γ_A and Γ_B are locally legal future-support domains. Detector settings are absent from $\mathcal{S}_{AB}$.

The shared pair-admissibility object

3.1 Native definition

Transport the B-side frame generator into the A-side source frame and define

$$\mathbf{J}_{\text{pair}} = \mathbf{J}_A + \text{Ad}_{[R_{AB}]}(\mathbf{J}_B). \quad (3.1)$$

The selected source channel requires

$$\mathbf{J}_{\text{pair}}^2 = 0. \quad (3.2)$$

Let Π_{legal}^{AB} be the orthogonal projector onto the source branch, port, support, history, and no-overwrite legal subspace. Define the legal zero-total-frame domain by the intersection

$$\Omega_{AB}^{(0)} = \{|\psi\rangle : \Pi_{\text{legal}}^{AB} |\psi\rangle = |\psi\rangle, \quad \mathbf{J}_{\text{pair}}^2 |\psi\rangle = 0\}. \quad (3.3)$$

Let Π_{Ω} denote the orthogonal projector onto this intersection.

If Π_{legal}^{AB} commutes with the zero-frame projector Π_0^{AB} , then

$$\Pi_{\Omega} = \Pi_{\text{legal}}^{AB} \Pi_0^{AB}. \quad (3.4)$$

Without that commutation, the sandwich

$$\Pi_{\text{legal}}^{AB} \Pi_0^{AB} \Pi_{\text{legal}}^{AB} \quad (3.5)$$

is a positive legal-support filter but is not automatically idempotent and must not be called a projector. Equation (3.3) is the canonical definition in either case.

The native object is a restricted domain of jointly admissible future pair histories carrying one selected source relation. It is not a new field, signal channel, readable hidden label, or counterfactual answer table.

3.2 Reduced two-state representation

Work in the common source frame after transporting the B-side generator. For two spin- $\frac{1}{2}$ closure supports in units $\hbar = 1$,

$$\mathbf{J}_{\text{tot}} = \mathbf{S}_A + \mathbf{S}_B = \frac{1}{2}(\boldsymbol{\sigma}_A + \boldsymbol{\sigma}_B). \quad (3.6)$$

Therefore

$$\mathbf{J}_{\text{tot}}^2 = \frac{1}{4}(\boldsymbol{\sigma}_A^2 + \boldsymbol{\sigma}_B^2 + 2\boldsymbol{\sigma}_A \cdot \boldsymbol{\sigma}_B) \quad (3.7)$$

$$= \frac{3}{2}\mathbb{I} + \frac{1}{2}\boldsymbol{\sigma}_A \cdot \boldsymbol{\sigma}_B. \quad (3.8)$$

Its eigenvalues are 0 on the singlet sector and 2 on the triplet sector. Hence

$$\Pi_0 = \mathbb{I} - \frac{1}{2}\mathbf{J}_{\text{tot}}^2 \quad (3.9)$$

$$= \frac{1}{4}(\mathbb{I} - \boldsymbol{\sigma}_A \cdot \boldsymbol{\sigma}_B). \quad (3.10)$$

Directly,

$$\Pi_0^2 = \Pi_0, \quad \text{rank } \Pi_0 = 1, \quad \text{Tr } \Pi_0 = 1. \quad (3.11)$$

For the selected legal source branch, the reduced representation of Π_Ω is Π_0 .

3.3 Nonfactorizability

A representative normalized vector is

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|0\rangle_A |1\rangle_B - |1\rangle_A |0\rangle_B), \quad \Pi_0 = |\Psi^-\rangle \langle \Psi^-|. \quad (3.12)$$

Its coefficient matrix has rank two, so the Schmidt rank is two. No product vector lies in $\text{Im}(\Pi_0)$ and no separable source mixture equals the pure rank-one projector. Thus the selected support is nonfactorizable. This establishes a source relation, not yet a Bell violation or probability law.

3.4 Local noncommitment

Taking either local restriction,

$$\text{Tr}_B \Pi_0 = \frac{1}{2}\mathbb{I}_A, \quad \text{Tr}_A \Pi_0 = \frac{1}{2}\mathbb{I}_B. \quad (3.13)$$

The source relation is definite while neither individual support carries an accessible preassigned orientation. This is stronger than ignorance about one hidden axis: the local restriction is invariant under every rotation.

3.5 Common rotation invariance

For $U \in SU(2)$,

$$(U \otimes U)\Pi_0(U^\dagger \otimes U^\dagger) = \Pi_0. \quad (3.14)$$

Equivalently,

$$(\mathbf{n} \cdot \boldsymbol{\sigma}_A + \mathbf{n} \cdot \boldsymbol{\sigma}_B)\Pi_0 = 0 \quad (3.15)$$

for every unit axis $\mathbf{n}$. The source annihilates total frame along every common analyzer axis.

3.6 Same-axis support exclusion

Let $P_s(\mathbf{n})$ be a sharp local sign projector, derived in the next chapter. The zero-total-frame condition implies

$$\Pi_0 [P_s(\mathbf{n}) \otimes P_s(\mathbf{n})] \Pi_0 = 0, \quad (3.16)$$

while the two opposite-sign sectors exhaust the source support. Rotational symmetry and local neutrality force equal support weight for $+-$ and $-+$.

M3–M4 pair theorem

Conditional on the selected pure $L = 0, S = 0$ source channel, the legal pair-support object is one-dimensional, common-rotation invariant, nonfactorizable, locally orientation-neutral, and excludes equal signs along every common sharp analyzer axis.

3.7 Measurement-independence and private-history firewalls

The source object contains H_v , the selected branch, retarded frame transports, the relative holonomy, and legality domains. It contains neither future settings $(\mathbf{a}, \mathbf{b})$ nor values for unrealized contexts. The statement

$$\mathbb{P}(\mathbf{a}, \mathbf{b} \mid \Lambda_{\text{src}}) = \mathbb{P}(\mathbf{a}, \mathbf{b}) \quad (3.17)$$

remains a separate premise and audit obligation; it does not follow merely because the source schema omits settings.

Local detector projectors and contextual readout

4.1 Detector problem

The source relation must not contain detector settings. The local analyzer law must therefore be derived independently from the two-state closure support and local apparatus symmetries.

Let $D(\mathbf{n})$ be the signed observable associated with one physical analyzer axis $\mathbf{n}$. The assumptions are:

1. $D(\mathbf{n})$ is Hermitian and acts only on the local two-state closure support;
2. rotational covariance: $D(R\mathbf{n}) = U(R)D(\mathbf{n})U(R)^\dagger$;
3. axis reversal: $D(-\mathbf{n}) = -D(\mathbf{n})$;
4. sharp binary closure: the ideal outcomes are $s = \pm 1$ and $D(\mathbf{n})^2 = \mathbb{I}$.

4.2 Uniqueness of the local signed analyzer

Every Hermitian 2×2 operator can be written

$$D(\mathbf{n}) = \alpha(\mathbf{n})\mathbb{I} + \boldsymbol{\beta}(\mathbf{n}) \cdot \boldsymbol{\sigma}. \quad (4.1)$$

Rotational covariance on the unit sphere implies α is constant and $\boldsymbol{\beta}$ is parallel to $\mathbf{n}$:

$$D(\mathbf{n}) = \alpha\mathbb{I} + \beta\mathbf{n} \cdot \boldsymbol{\sigma}. \quad (4.2)$$

Axis reversal forces $\alpha = 0$. Sharpness gives

$$D(\mathbf{n})^2 = \beta^2\mathbb{I} = \mathbb{I}, \quad (4.3)$$

so $|\beta| = 1$. A global sign is a detector-label convention. Choosing the positive calibration,

$$D(\mathbf{n}) = \mathbf{n} \cdot \boldsymbol{\sigma}. \quad (4.4)$$

The local sign projectors are

$$P_s(\mathbf{n}) = \frac{1}{2} (\mathbb{I} + s \mathbf{n} \cdot \boldsymbol{\sigma}), \quad s \in \{-1, +1\}. \quad (4.5)$$

They obey

$$P_s P_{s'} = \delta_{ss'} P_s, \quad \sum_s P_s = \mathbb{I}. \quad (4.6)$$

4.3 Why higher axis powers do not create a new law

Because

$$(\mathbf{n} \cdot \boldsymbol{\sigma})^2 = \mathbb{I}, \quad (4.7)$$

the spectral calculus of any function defined on the two eigenvalues $\{-1, +1\}$ reduces exactly to

$$f(\mathbf{n} \cdot \boldsymbol{\sigma}) = c_0 \mathbb{I} + c_1 \mathbf{n} \cdot \boldsymbol{\sigma}. \quad (4.8)$$

Analyticity is not required. Axis reversal removes the identity term from a signed detector observable. The linear generator is therefore the exact algebraic form within the two-state support, not a leading-order approximation.

4.4 Calibrated unsharp extension

A symmetric local loss of analyzer contrast is represented by the effects

$$E_s(\mathbf{n}) = \frac{1}{2} (\mathbb{I} + s\eta \mathbf{n} \cdot \boldsymbol{\sigma}), \quad 0 \leq \eta \leq 1. \quad (4.9)$$

Positivity follows because the eigenvalues are $(1 \pm \eta)/2$.

For $\eta < 1$, the E_s are POVM effects rather than orthogonal projectors. They describe a downstream calibrated response channel and must not be substituted into the sharp-branch orthogonality proof without that distinction.

4.5 Locality of the analyzer map

On the pair support,

$$P_s^A(\mathbf{a}) = P_s(\mathbf{a}) \otimes \mathbb{I}_B, \quad P_t^B(\mathbf{b}) = \mathbb{I}_A \otimes P_t(\mathbf{b}). \quad (4.10)$$

Hence

$$[P_s^A(\mathbf{a}), P_t^B(\mathbf{b})] = 0. \quad (4.11)$$

The setting $\mathbf{a}$ appears only in the A-side map and $\mathbf{b}$ only in the B-side map.

Detector microphysics

The projector law is a conditional theorem for an ideal calibrated two-state analyzer. The microscopic interaction U_{det} , irreversible amplification, detector-material latency, saturation, and objective macroscopic record formation remain physical detector debts.

Joint support composition and the angular law

5.1 Actual-context joint sectors

For one physically selected detector context $(\mathbf{a}, \mathbf{b})$ define

$$Q_{st}(\mathbf{a}, \mathbf{b}) = P_s^A(\mathbf{a})P_t^B(\mathbf{b}) = P_s(\mathbf{a}) \otimes P_t(\mathbf{b}). \quad (5.1)$$

The four sharp Q_{st} are mutually orthogonal and exhaustive:

$$Q_{st}Q_{s't'} = \delta_{ss'}\delta_{tt'}Q_{st}, \quad \sum_{s,t} Q_{st} = \mathbb{I}_{AB}. \quad (5.2)$$

The candidate branch map on the selected source support is

$$C_{st}(\mathbf{a}, \mathbf{b}) = Q_{st}(\mathbf{a}, \mathbf{b})\Pi_0. \quad (5.3)$$

Distinct branches are orthogonal in squared-support geometry:

$$C_{st}^\dagger C_{s't'} = \Pi_0 Q_{st} Q_{s't'} \Pi_0 = 0 \quad \text{for } (s, t) \neq (s', t'). \quad (5.4)$$

5.2 Representation support measure

Define

$$\mu_{st}(\mathbf{a}, \mathbf{b}) = \text{Tr}(C_{st}^\dagger C_{st}) = \text{Tr}(\Pi_0 Q_{st}). \quad (5.5)$$

At this stage μ_{st} is not yet a probability or settlement intensity. It is the squared amount of selected source support that survives into one legal sharp candidate sector.

5.3 Direct trace derivation

Insert

$$\Pi_0 = \frac{1}{4} \left(\mathbb{I} - \sum_{i=1}^3 \sigma_i \otimes \sigma_i \right) \quad (5.6)$$

and

$$Q_{st} = \frac{1}{4} (\mathbb{I} + s \mathbf{a} \cdot \boldsymbol{\sigma}) \otimes (\mathbb{I} + t \mathbf{b} \cdot \boldsymbol{\sigma}). \quad (5.7)$$

Then

$$\mu_{st} = \frac{1}{16} \text{Tr} \left[\left(\mathbb{I} - \sum_i \sigma_i \otimes \sigma_i \right) (\mathbb{I} + s a_j \sigma_j) \otimes (\mathbb{I} + t b_k \sigma_k) \right]. \quad (5.8)$$

Use

$$\text{Tr} \sigma_i = 0, \quad \text{Tr}(\sigma_i \sigma_j) = 2\delta_{ij}. \quad (5.9)$$

The identity contribution is $4/16 = 1/4$. Every term linear in one Pauli matrix vanishes. The only nonzero correlation term is

$$-\frac{st}{16} \sum_{i,j,k} a_j b_k \text{Tr}(\sigma_i \sigma_j) \text{Tr}(\sigma_i \sigma_k) = -\frac{st}{4} \sum_i a_i b_i \quad (5.10)$$

$$= -\frac{st}{4} \mathbf{a} \cdot \mathbf{b}. \quad (5.11)$$

Therefore

$$\boxed{\mu_{st}(\mathbf{a}, \mathbf{b}) = \frac{1}{4} [1 - st \mathbf{a} \cdot \mathbf{b}].} \quad (5.12)$$

This is the first point at which the familiar angular numerator is allowed to appear. It follows from the independently fixed source and detector projectors.

5.4 Geometric checks

For unit axes,

$$0 \leq \mu_{st} \leq \frac{1}{2}, \quad \sum_{s,t} \mu_{st} = 1. \quad (5.13)$$

The regressions are

$$\mathbf{a} = \mathbf{b} : \quad \mu_{++} = \mu_{--} = 0, \quad \mu_{+-} = \mu_{-+} = \frac{1}{2}; \quad (5.14)$$

$$\mathbf{a} = -\mathbf{b} : \quad \mu_{+-} = \mu_{-+} = 0, \quad \mu_{++} = \mu_{--} = \frac{1}{2}; \quad (5.15)$$

$$\mathbf{a} \cdot \mathbf{b} = 0 : \quad \mu_{st} = \frac{1}{4} \quad \text{for all } s, t. \quad (5.16)$$

5.5 Support-level no-signaling cancellation

Summing over the remote sign,

$$\sum_t \mu_{st}(\mathbf{a}, \mathbf{b}) = \frac{1}{4} \sum_t (1 - st \mathbf{a} \cdot \mathbf{b}) \quad (5.17)$$

$$= \frac{1}{2}, \quad (5.18)$$

independent of $\mathbf{b}$. Similarly,

$$\sum_s \mu_{st}(\mathbf{a}, \mathbf{b}) = \frac{1}{2}, \quad (5.19)$$

independent of $\mathbf{a}$.

5.6 Unsharp-response diagnostic

Replacing the local projectors by the calibrated effects E_s gives

$$w_{st}^{(\eta_A, \eta_B)} = \text{Tr}[\Pi_0(E_s^A(\mathbf{a}) \otimes E_t^B(\mathbf{b}))] = \frac{1}{4} [1 - st \eta_A \eta_B \mathbf{a} \cdot \mathbf{b}]. \quad (5.20)$$

These w_{st} are valid POVM weights. For $\eta < 1$ they are not generated by four mutually orthogonal projectors, so this extension is a detector-response diagnostic rather than part of the sharp M6 orthogonality theorem.

M6 angular-support theorem

For sharp local analyzers, the four actual-context support sectors are positive, orthogonal, exhaustive, and have exact measure $\frac{1}{4}(1 - st \mathbf{a} \cdot \mathbf{b})$. The angular law is derived at support level; an intensity and completed-event interpretation still has to be earned.

Contextual completion and the native intensity bridge

6.1 M7: one actual context per run

Let the local apparatus setting registers have orthogonal projectors K_a^A and K_b^B . Define

$$K_{ab} = K_a^A \otimes K_b^B. \quad (6.1)$$

A controlled detector interaction may be written

$$U_{\text{ctrl}} = \sum_{a,b} K_{ab} \otimes U_{ab}. \quad (6.2)$$

In one real run, exactly one setting-register block is activated. The source object remains setting free; the selected detector block supplies the local projectors for the performed context only.

The mathematical projectors for unrealized contexts remain useful for comparing possible experiments, but they have zero activated apparatus support in the actual run. The physical ledger therefore need not contain a global Boolean refinement assigning outcomes to all settings simultaneously.

6.2 One-hot record algebra

For the actual context (a, b) , the four candidate outcome sectors are mutually exclusive. An ideal successful record algebra has one-hot form

$$R_{st} R_{s't'} = \delta_{ss'} \delta_{tt'} R_{st}, \quad \sum_{s,t} R_{st} = \mathbb{I}_{\text{success}}. \quad (6.3)$$

At most one completed record is earned by exclusivity. At least one completed record requires an explicit completion premise; nonideal trials require null or abort sectors.

6.3 M8: from support measure to relative intensity

Let an eligible orthogonal branch have support measure $\mu \geq 0$. Let $f(\mu)$ be the relative settlement/completion intensity assigned to that branch.

6.3.1 Positivity and orthogonal additivity

Assume

$$f(\mu) \geq 0 \quad \text{for } \mu \geq 0, \quad (6.4)$$

and for orthogonal refinements,

$$f(\mu_1 + \mu_2) = f(\mu_1) + f(\mu_2). \quad (6.5)$$

Linearity lemma

Set $\kappa = f(1)$. Additivity gives $f(0) = 0$ and makes f monotone on $\mathbb{R}_+$: if $y \geq x$, then $f(y) = f(x) + f(y - x) \geq f(x)$. For positive integers m, n ,

$$f\left(\frac{m}{n}\right) = \frac{m}{n} f(1) = \kappa \frac{m}{n}. \quad (6.6)$$

For any real $x \geq 0$, choose rational sequences $r_n \uparrow x$ and $q_n \downarrow x$. Monotonicity gives

$$\kappa r_n = f(r_n) \leq f(x) \leq f(q_n) = \kappa q_n. \quad (6.7)$$

Taking the limit yields

$$\boxed{f(x) = \kappa x, \quad \kappa \geq 0.} \quad (6.8)$$

Thus nonnegative orthogonal additivity already excludes pathological Cauchy solutions. Equal-cell refinement invariance follows automatically.

For n equal orthogonal cells,

$$f(\mu) = n f\left(\frac{\mu}{n}\right). \quad (6.9)$$

Nonlinear candidates such as $f(\mu) = \mu^p$, $p \neq 1$, fail:

$$n \left(\frac{\mu}{n}\right)^p = n^{1-p} \mu^p \neq \mu^p. \quad (6.10)$$

6.4 Calibrated channel symmetry

Independent sign relabelings are detector conventions. The ideal analyzer class therefore includes the calibration premise that the four sign sectors of one actual context share one common positive completion

scale. Write

$$I_{st}(\mathbf{a}, \mathbf{b}) = \kappa_{ab} \mu_{st}(\mathbf{a}, \mathbf{b}), \quad (6.11)$$

so

$$I_{st}(\mathbf{a}, \mathbf{b}) = \frac{\kappa_{ab}}{4} [1 - st \mathbf{a} \cdot \mathbf{b}]. \quad (6.12)$$

The common factor is only a relative scale within the selected ideal context. It cannot be used as a hidden pair-setting-dependent success filter. Any nonideal setting-dependent success process must be represented later by explicit normalized local response channels and null records.

6.5 Completion and probability

Suppose eligible microscopic completion attempts are additive and have relative hazards I_{st} . The distribution over final joint record types is

$$\mathbb{P}(s, t \mid \mathbf{a}, \mathbf{b}, \text{ideal completion}) = \frac{I_{st}}{\sum_{r,q} I_{rq}}. \quad (6.13)$$

Because $\sum_{r,q} \mu_{rq} = 1$,

$$\mathbb{P}(s, t \mid \mathbf{a}, \mathbf{b}) = \mu_{st} = \frac{1}{4} [1 - st \mathbf{a} \cdot \mathbf{b}]. \quad (6.14)$$

The phrase “completion race” refers to weights over final record types. It is not a coordinate-time race that chooses which spacelike detector physically acts first; M9 and H5 make that boundary explicit.

6.6 Correlation and marginals

The local marginals are

$$\mathbb{P}_A(s \mid \mathbf{a}, \mathbf{b}) = \frac{1}{2}, \quad \mathbb{P}_B(t \mid \mathbf{a}, \mathbf{b}) = \frac{1}{2}. \quad (6.15)$$

The product correlation is

$$E(\mathbf{a}, \mathbf{b}) = \sum_{s,t} st \mathbb{P}(s, t \mid \mathbf{a}, \mathbf{b}) \quad (6.16)$$

$$= -\mathbf{a} \cdot \mathbf{b}. \quad (6.17)$$

Same-axis perfect anticorrelation and unbiased local outcomes are recovered without a preassigned orientation table.

6.7 Loss extension

Outcome-independent local efficiencies multiply all four successful sectors by a common local product,

$$I_{st}^{\text{click}} = \eta_A(\mathbf{a})\eta_B(\mathbf{b})I_{st}. \quad (6.18)$$

Conditioning on joint success then preserves the sign kernel, but the conditioning is not by itself a loophole-free all-trial Bell test. Outcome-dependent efficiencies do not generally factor out and must be represented by explicit null channels.

M7–M8 completion theorem

One actual detector context activates four orthogonal sharp candidate sectors. Nonnegative orthogonal additivity forces relative intensity to be linear in the M6 measure. Under ideal calibrated completion, normalization yields the exact singlet joint law and $E = -\mathbf{a} \cdot \mathbf{b}$ without inserting the target kernel.

Absolute completion debt

The derivation fixes relative intensities. It does not derive the absolute scale κ_{ab} , microscopic settlement attempt rate, detector amplification time, or material efficiency. Those require source/detector dynamics and experimental calibration.

Order-independent records, no-signaling, and measurement independence

7.1 M9: local branch-record maps

Use the tensor order

$$\mathcal{H}_{R_A} \otimes \mathcal{H}_{R_B} \otimes \mathcal{H}_A \otimes \mathcal{H}_B. \quad (7.1)$$

Let R_A, R_B have ready states $|0\rangle_{R_A}, |0\rangle_{R_B}$. Ideal branch Kraus maps may be represented as

$$W_A^s(\mathbf{a}) = |s\rangle\langle 0|_{R_A} \otimes \mathbb{I}_{R_B} \otimes P_s(\mathbf{a}) \otimes \mathbb{I}_B, \quad (7.2)$$

$$W_B^t(\mathbf{b}) = \mathbb{I}_{R_A} \otimes |t\rangle\langle 0|_{R_B} \otimes \mathbb{I}_A \otimes P_t(\mathbf{b}). \quad (7.3)$$

The sums over all local branches, together with any required null branch, define the complete local instruments. For every pair of branches, the maps act on disjoint factors, so

$$\boxed{W_A^s(\mathbf{a})W_B^t(\mathbf{b}) = W_B^t(\mathbf{b})W_A^s(\mathbf{a}).} \quad (7.4)$$

A-first and B-first coordinate narratives therefore produce the same completed branch vector and probability.

7.2 Ledger partial order

The causal record structure is a partial order:

$$H_v \prec H_A^s, \quad H_v \prec H_B^t, \quad (7.5)$$

with no invariant order between spacelike H_A^s and H_B^t . Their common-future join is

$$H_C = H_v \sqcup H_A^s \sqcup H_B^t. \quad (7.6)$$

Associativity and commutativity of the join encode one final record object without a preferred first detector.

7.3 Sequential conditional descriptions

The invariant joint law can be factored in either narrative:

$$\mathbb{P}(s, t \mid a, b) = \mathbb{P}_A(s \mid a) \mathbb{P}_B(t \mid s, a, b) = \mathbb{P}_B(t \mid b) \mathbb{P}_A(s \mid t, a, b). \quad (7.7)$$

These are factorizations of one joint distribution, not physical remote updates. Neither wing's local marginal changes with the remote setting.

7.4 C5: arbitrary local null-complete response channels

Let ideal signs be processed through normalized local stochastic channels with explicit nulls:

$$D_A(\alpha \mid s, a), \quad D_B(\beta \mid t, b), \quad (7.8)$$

where α, β may include $+, -, 0$, misclassification, saturation, or latency bins. The observed all-trial law is

$$q(\alpha, \beta \mid a, b) = \sum_{s,t} D_A(\alpha \mid s, a) D_B(\beta \mid t, b) p_{st}(a, b). \quad (7.9)$$

Normalization on the B side gives

$$q_A(\alpha \mid a, b) = \sum_{\beta} q(\alpha, \beta \mid a, b) \quad (7.10)$$

$$= \sum_s D_A(\alpha \mid s, a) \sum_t p_{st}(a, b) \quad (7.11)$$

$$= \frac{1}{2} \sum_s D_A(\alpha \mid s, a), \quad (7.12)$$

which is independent of b . The B result is symmetric.

Null-complete no-signaling theorem

Every normalized local stochastic response channel preserves unconditional remote-setting independence when the input law is no-signaling and all local records, including nulls, are retained.

7.5 Why selected samples are different

A selected sample conditioned on a joint event $K(\alpha, \beta, a, b)$ has

$$q_K(\alpha, \beta \mid a, b) = \frac{K(\alpha, \beta, a, b)q(\alpha, \beta \mid a, b)}{\sum_{\alpha', \beta'} K(\alpha', \beta', a, b)q(\alpha', \beta' \mid a, b)}. \quad (7.13)$$

The denominator is setting-pair dependent in general. Coincidence, timing, order, or outcome-conditioned samples need not preserve no-signaling even when the raw all-trial law does.

7.6 C6: measurement independence

Let Λ_{src} contain the complete pre-setting source-side information: source history, geometry, timing, intensity, impurities, drift, run block, and diagnostics. Measurement independence is

$$\boxed{P(a, b \mid \Lambda_{\text{src}}) = P(a, b)}. \quad (7.14)$$

This is an explicit premise and empirical audit target, not a theorem of schema separation. Late or spacelike setting generation blocks ordinary source-to-setting causal influence but cannot logically exclude every common-cause or superdeterministic model.

Finite data can bound deviations using mutual information, setting-balance tests, classifiers, lag scans, block stability, and filter-stage re-audits. It cannot prove exact equality for all hidden variables.

7.7 Construction result and Bell-test entry

Combining M1–M9 gives the ideal completed sign law independently of any Bell-local hidden-variable comparison:

SFT ideal record theorem

Conditional on the selected pure $L = 0$, $S = 0$ local source channel, ideal calibrated local two-state analyzers, nonnegative additive branch completion, and commuting local record maps,

$$\mathbb{P}(s, t \mid \mathbf{a}, \mathbf{b}) = \frac{1}{4} (1 - st \mathbf{a} \cdot \mathbf{b}). \quad (7.15)$$

The local marginals are unbiased and no-signaling, same-axis outcomes are perfectly anticorrelated, and no preferred spacelike measurement order or superluminal physical update is required.

Measurement independence is added only when the distribution is tested against the Bell-local class. A complete all-trial protocol is added only when the mathematical result is used for laboratory inference.

The next chapter keeps those logical layers separate.

Hostile closure I: Bell locality, CHSH, detectors, and measurement dependence

8.1 H1: Bell-local factorization kill

A measurement-independent Bell-local model has

$$\mathbb{P}(s, t \mid a, b) = \int d\lambda \rho(\lambda) \mathbb{P}_A(s \mid a, \lambda) \mathbb{P}_B(t \mid b, \lambda), \quad (8.1)$$

with $\rho(\lambda) \geq 0$, $\int \rho = 1$, and no remote setting in either local response. Every stochastic local response can be refined by adding local auxiliary randomness, so the extreme points are deterministic tables

$$A_x(\lambda), B_y(\lambda) \in \{-1, +1\}. \quad (8.2)$$

Perfect same-axis anticorrelation gives

$$\mathbb{P}(A_x = B_x) = 0. \quad (8.3)$$

Because the factored integrand is nonnegative, this forces

$$B_x(\lambda) = -A_x(\lambda) \quad (8.4)$$

for almost every contributing deterministic refinement.

For three axes a, b, c , deterministic signs obey

$$|A_a A_b - A_a A_c| = |A_b - A_c| = 1 - A_b A_c. \quad (8.5)$$

Averaging and using $B = -A$ gives Bell's 1964 bound [1]

$$\boxed{|E(a, b) - E(a, c)| \leq 1 + E(b, c)}. \quad (8.6)$$

Choose coplanar angles 0° , 60° , and 120° . The frozen law gives

$$E(a, b) = -\frac{1}{2}, \quad E(a, c) = +\frac{1}{2}, \quad E(b, c) = -\frac{1}{2}. \quad (8.7)$$

The local bound would require $1 \leq 1/2$, an exact contradiction margin $1/2$.

A full 3×3 local-polytope linear program is infeasible. The nearest nonnegative local model has minimum maximum probability-cell residual $1/32$. An exact signed strategy representation requires negative mass $1/8$, which is not an ordinary probability model.

H1 conclusion

No setting-independent, nonnegative, normalized Bell-local factorization reproduces the frozen probability table. The observed distribution remains no-signaling; what fails is Bell-local factorizability under the stated premise.

8.2 H2: fixed CHSH witness and global maximum

Freeze the settings

$$\mathbf{a}_0 = (1, 0), \quad \mathbf{a}_1 = (0, 1), \quad \mathbf{b}_0 = \frac{(1, 1)}{\sqrt{2}}, \quad \mathbf{b}_1 = \frac{(1, -1)}{\sqrt{2}}. \quad (8.8)$$

For the CHSH combination [2]

$$S = E_{00} + E_{01} + E_{10} - E_{11}, \quad (8.9)$$

the frozen correlations are

$$E_{00} = E_{01} = E_{10} = -\frac{1}{\sqrt{2}}, \quad E_{11} = +\frac{1}{\sqrt{2}}, \quad (8.10)$$

so

$$S = -2\sqrt{2}, \quad |S| = 2\sqrt{2}. \quad (8.11)$$

Every deterministic local table has $|S| = 2$, so every nonnegative local mixture satisfies $|S| \leq 2$.

The fixed witness is also the global geometric maximum of $E = -\mathbf{a} \cdot \mathbf{b}$:

$$|S| = |\mathbf{a}_0 \cdot (\mathbf{b}_0 + \mathbf{b}_1) + \mathbf{a}_1 \cdot (\mathbf{b}_0 - \mathbf{b}_1)| \quad (8.12)$$

$$\leq \|\mathbf{b}_0 + \mathbf{b}_1\| + \|\mathbf{b}_0 - \mathbf{b}_1\| \quad (8.13)$$

$$\leq 2\sqrt{2}. \quad (8.14)$$

The predeclared axes saturate both inequalities.

For visibility V ,

$$|S| = 2\sqrt{2}V, \quad (8.15)$$

so violation requires $V > 1/\sqrt{2}$. For equal local analyzer sharpness η with $V = 1$,

$$\eta > 2^{-1/4}. \quad (8.16)$$

8.3 H3: arbitrary local detector processing

Let each ideal sign be mapped locally into a ternary record $-1, 0, +1$. A deterministic map from one binary input sign to one ternary output has $3^2 = 9$ possibilities for each setting. With four local settings there are

$$9^4 = 6561 \quad (8.17)$$

deterministic local postprocessing networks. Exhaustive enumeration found

$$\max |S_{\text{raw}}| = 2\sqrt{2}, \quad \max \text{ raw no-signaling error} = 0. \quad (8.18)$$

Every stochastic local channel is a convex mixture of deterministic channels, so no normalized local processing can increase the raw frozen CHSH maximum.

For unbiased signed gains g_A, g_B and source visibility V ,

$$|S_{\text{raw}}| = 2\sqrt{2}Vg_Ag_B, \quad (8.19)$$

and violation requires $Vg_Ag_B > 1/\sqrt{2}$.

8.3.1 Postselection counterexample

Outcome-dependent local success rates may be written

$$\eta_A(s|x) = \bar{\eta}_A(x)(1 + s\delta_{Ax}), \quad \eta_B(t|y) = \bar{\eta}_B(y)(1 + t\delta_{By}). \quad (8.20)$$

After coincidence conditioning, each context is renormalized by a different denominator. The explicit family

$$\delta_{A0} = 0, \quad \delta_{A1} = -d, \quad \delta_{B0} = d, \quad \delta_{B1} = -d \quad (8.21)$$

gives

$$|S_{\text{cond}}(d)| = \sqrt{2} + 2 \frac{1/\sqrt{2} + d^2}{1 + d^2/\sqrt{2}}. \quad (8.22)$$

For every $d \neq 0$, $|S_{\text{cond}}| > 2\sqrt{2}$, and as $d \rightarrow 1$,

$$|S_{\text{cond}}| \rightarrow 2 + \sqrt{2}. \quad (8.23)$$

The selected A-side signed expectation can differ between remote settings by an amount approaching $\sqrt{2}$; equivalently, the selected $A = +$ probability can shift by $1/\sqrt{2}$. These are not raw signals. They are comparisons of differently renormalized subensembles and do not define one normalized no-signaling kernel.

Timing windows and A-before-B selections produce the same class of artifact. Raw null-complete records remain no-signaling.

8.4 H4: exact measurement-dependence escape

Relax only measurement independence:

$$q_{xy}(\lambda) = \mathbb{P}(\lambda \mid x, y). \quad (8.24)$$

Keep deterministic local response tables, nonnegative normalized weights, the frozen settings, and the complete detector contract.

Among the 16 deterministic binary tables, select the eight that fail exactly one of the four target CHSH product signs. Let

$$d = \frac{1 - 1/\sqrt{2}}{2}. \quad (8.25)$$

For actual context c , assign each of the two strategies that fail c weight $d/2$, and each of the other six weight $(1 - d)/6$. This reproduces all sixteen frozen joint probabilities and unbiased local marginals exactly.

The minimum possible maximum pairwise source-distribution total variation is

$$M_{\text{TV}}^{\min} = \max_{c, c'} \text{TV}(q_c, q_{c'}) = \frac{\sqrt{2} - 1}{3}. \quad (8.26)$$

The minimum average distance to one setting-independent distribution is

$$D_{\text{avg}}^{\min} = \min_r \frac{1}{4} \sum_c \text{TV}(q_c, r) = \frac{\sqrt{2} - 1}{4}. \quad (8.27)$$

If dependence is allowed on only one wing's setting,

$$M_{\text{one-wing}}^{\min} = \sqrt{2} - 1. \quad (8.28)$$

For uniformly sampled contexts, let H_2 denote binary entropy. The exact minimum mutual information is

$$I_{\min}(\Lambda; X, Y) = 2 - H_2(d) - (1 - d) \log_2 3 \approx 0.04627384685 \text{ bits}. \quad (8.29)$$

Why the information optimum is exact

A deterministic CHSH product-sign pattern must disagree with the target in an odd number of contexts: one or three. Mutual information is convex in the setting-conditioned source channel. Averaging any feasible channel over the symmetry group of the four frozen contexts cannot increase $I(\Lambda; X, Y)$ and reduces the problem to the one-failure and three-failure orbits. For the required mismatch rate $d < 1/4$, the three-failure orbit carries strictly more setting information. The optimum therefore uses only one-failure labels. Its symmetric channel has probability d for the label equal to the actual context and $(1 - d)/3$ for each other label, giving the stated rate-distortion expression. The global output-flip partner is independent of the context and contributes zero additional mutual information.

The optimal setting-pair guess probability rises only from $1/4$ to

$$\mathbb{P}_{\text{guess}} = \frac{1 - d}{3} \approx 0.2845177969. \quad (8.30)$$

For a visibility-scaled witness,

$$M_{\text{TV}}^{\min}(V) = \max \left[0, \frac{2\sqrt{2}V - 2}{6} \right]. \quad (8.31)$$

H1–H4 combined meaning

With measurement independence, the frozen law lies outside the Bell-local polytope. If measurement independence is deliberately relaxed, an exact detector-local nonnegative model exists, and the required dependence can be quantified. H4 does not supply evidence that such dependence occurs physically.

Hostile closure II: spacetime order, delayed choice, swapping, and GHZ

9.1 H5: relativistic frame-order variants

Generalize the local record maps to latency-resolved instruments

$$K_A(\alpha, \tau_A \mid s, a, \xi_A), \quad K_B(\beta, \tau_B \mid t, b, \xi_B), \quad (9.1)$$

where records may include sign, null status, timing, apparatus state, and local calibration variables. Because the two instruments act on disjoint tensor factors,

$$[K_A, K_B] = 0 \quad (9.2)$$

for every pair of local outcomes and latencies.

The complete latency-resolved law is

$$q(\alpha, \tau_A, \beta, \tau_B \mid a, b) = \sum_{s,t} D_A(\alpha, \tau_A \mid s, a) D_B(\beta, \tau_B \mid t, b) p_{st}(a, b). \quad (9.3)$$

Summing every remote record eliminates the remote channel, so raw no-signaling survives arbitrary local delay physics.

For spacelike detector events,

$$\Delta x^2 - c^2 \Delta t^2 > 0, \quad (9.4)$$

and a Lorentz boost may reverse the sign of Δt . Therefore no frame-invariant physical first detector exists. A rule that assigns different final records according to $\text{sgn } \Delta t_{\text{frame}}$ is inconsistent: two inertial descriptions of one event pair would produce different ledger histories.

The invariant object is the future join

$$H_C = H_v \sqcup H_A^{(\alpha, \tau_A)} \sqcup H_B^{(\beta, \tau_B)}. \quad (9.5)$$

Order-conditioned samples are unsafe. An explicit local latency model produced

$$\mathbb{P}(A = + \mid \tau_A < \tau_B, \text{ same axes}) = 1, \quad (9.6)$$

while

$$\mathbb{P}(A = + \mid \tau_A < \tau_B, \text{ orthogonal axes}) = 0, \quad (9.7)$$

even though both raw marginals were $1/2$. The unit apparent remote shift is a joint order-selection artifact.

9.2 H6: delayed-choice source/context ordering

Let S denote source formation, C_A, C_B the local setting writes, and D_A, D_B the local detector interactions. The required causal relations are

$$S \prec C_A \prec D_A, \quad S \prec C_B \prec D_B. \quad (9.8)$$

There are six linear extensions of C_A, D_A, C_B, D_B consistent with those chains. Because cross-wing controlled maps act on disjoint factors, all six yield the same final branch.

A local choice may occur later than the remote detector in some frames when the events are spacelike. This requires no update because the remote marginal is setting independent. A setting chosen after its own completed local record is impossible without history rewriting. If A 's outcome were complete before x existed, then for each fixed y ,

$$E(0, y) = E(1, y). \quad (9.9)$$

The frozen witness instead has

$$E(0, 1) = -\frac{1}{\sqrt{2}}, \quad E(1, 1) = +\frac{1}{\sqrt{2}}, \quad (9.10)$$

an exact contradiction $\sqrt{2}$.

Thus delayed choice means after source formation and possibly after the remote spacelike record in some frames, but always before the corresponding local analyzer interaction.

9.3 H7: entanglement swapping and later relational joins

Begin with two independent source pairs

$$\Omega_{12} \otimes \Omega_{34}, \quad (9.11)$$

with no initial Ω_{14} . For two initial singlets,

$$|\Psi^-\rangle_{12} \otimes |\Psi^-\rangle_{34}, \quad (9.12)$$

the unconditioned outer state is $\rho_{14} = \mathbb{I}_4/4$, with zero negativity.

Particles 2 and 3 meet locally at a Bell-sector analyzer with projector Π_μ^{23} . Let $\hat{\Pi}_\mu^{23}$ denote its embedding into the ordered four-particle space $\mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \mathcal{H}_3 \otimes \mathcal{H}_4$. The middle record writes

$$J_\mu = H_{12} \sqcup H_{34} \sqcup H_M^\mu. \quad (9.13)$$

The heralded outer restriction is

$$\rho_{14|\mu} = \frac{\text{Tr}_{23}[\hat{\Pi}_\mu^{23} \rho_{1234} \hat{\Pi}_\mu^{23}]}{\mathbb{P}(\mu)}. \quad (9.14)$$

Each middle outcome has probability 1/4 and maps to the corresponding outer Bell sector, up to global phase. Every conditioned outer state has Schmidt rank two and negativity 1/2. Ignoring the middle record gives

$$\sum_\mu \mathbb{P}(\mu) \rho_{14|\mu} = \frac{1}{4} \mathbb{I}_4. \quad (9.15)$$

This is the standard swapping identity [11]; the delayed-order stress mirrors the later delayed-choice implementation target [12].

Outer projectors act on systems 1 and 4, while the middle projector acts on 2 and 3. They commute, so middle-first, outer-first, and mixed descriptions give the same complete joint weights. A Bell analyzer yields entangled conditioned outer subensembles; a product analyzer yields separable conditioned subensembles. The unconditioned outer state is the same in both cases.

H7 ontology upgrade

A universal birth-only account of entanglement is insufficient. A later local joint interaction can join previously independent source ancestries and write a heralded relational ledger sector. This does not propagate an instantaneous physical boundary between distant outer particles and does not rewrite their earlier histories.

The classical middle tag is required to identify or correct the outer Bell sector. Without it, no usable swapped relation or signal exists.

9.4 H8: GHZ and genuine N -party support

Consider the selected three-support source

$$|\text{GHZ}^+\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle), \quad \Pi_{\text{GHZ}} = |\text{GHZ}^+\rangle\langle\text{GHZ}^+|. \quad (9.16)$$

Its Schmidt rank is two across $A|BC$, $B|AC$, and $C|AB$. Yet every pairwise reduced state is

$$\rho_{AB} = \rho_{AC} = \rho_{BC} = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|), \quad (9.17)$$

with zero negativity. The completed global coherent source state cannot therefore be reconstructed from its pairwise reduced states alone.

For one actual three-party context,

$$Q_{stu}(a, b, c) = P_s^A(a) \otimes P_t^B(b) \otimes P_u^C(c), \quad (9.18)$$

and

$$\mu_{stu} = \text{Tr}(\Pi_{\text{GHZ}} Q_{stu}). \quad (9.19)$$

The same nonnegative additive refinement argument conditionally extends the relative intensity law:

$$I_{stu} = \kappa_{abc} \mu_{stu}, \quad \mathbb{P}(stu | abc) = \mu_{stu}. \quad (9.20)$$

This is architectural consistency conditional on the prepared GHZ source sector; it is not a derivation of the native source generator U_{GHZ} .

For the canonical contexts [9, 10],

$$XXX = +1, \quad XYY = YXY = YYX = -1. \quad (9.21)$$

In each context, four sign triples with the required product parity have probability 1/4 and the other four have zero support. Every one-party marginal is 1/2 and every two-party marginal is 1/4, independent of the third setting.

A deterministic Bell-local table would require

$$x_A y_B y_C = -1, \quad (9.22)$$

$$y_A x_B y_C = -1, \quad (9.23)$$

$$y_A y_B x_C = -1, \quad (9.24)$$

$$x_A x_B x_C = +1. \quad (9.25)$$

Multiplying the first three gives $x_A x_B x_C = -1$, contradicting the fourth. None of the 64 deterministic tables

survives.

Define

$$M = \langle XXX \rangle - \langle XYY \rangle - \langle YXY \rangle - \langle YYX \rangle. \quad (9.26)$$

Every deterministic Bell-local table has $|M| = 2$, while the GHZ support gives

$$\boxed{M = 4.} \quad (9.27)$$

The full 32-cell local-polytope LP is infeasible; the nearest local model misses at least one cell by $1/16$, and an exact signed representation requires negative mass $1/2$.

H8 ontology upgrade

A static relational state description cannot be universally reduced to pairwise reduced objects. The SFT ontology must admit genuine N -party support objects for prepared source sectors whose global coherence is absent from every pairwise restriction. This does not forbid pairwise local interactions from dynamically preparing such a state.

H7–H8 microscopic debts

The projector algebra closes, but first-principles middle-analyzer dynamics, mode matching, GHZ source selection, U_{GHZ} , and material amplification remain open.

Hostile closure III: decoherence, impurity, loss, and the final detection attack

10.1 H9: external-history overlap as decoherence variable

Write two relational alternatives correlated with external history supports:

$$|0\rangle|e_0\rangle, \quad |1\rangle|e_1\rangle. \quad (10.1)$$

After those external records are not retained as part of the measured support, the surviving coherence is controlled by

$$\boxed{\gamma = \langle e_1|e_0\rangle, \quad |\gamma| \leq 1.} \quad (10.2)$$

The phase and modulus have different meanings:

- $|\gamma| = 1$ with $\arg \gamma \neq 0$ is coherent phase/frame drift;
- $|\gamma| < 1$ is loss of relational overlap to distinguishable external history;
- $\gamma = 0$ is complete dephasing.

No new field is introduced; γ summarizes overlap between existing environmental/history supports.

10.2 Noisy Bell family

Define

$$\rho_B(\gamma) = \frac{1}{2}(|01\rangle\langle 01| + |10\rangle\langle 10| - \gamma|01\rangle\langle 10| - \gamma^*|10\rangle\langle 01|), \quad (10.3)$$

and the isotropically contaminated family

$$\rho_B(V, \gamma) = V\rho_B(\gamma) + (1 - V)\frac{1}{4}\mathbb{I}_4, \quad 0 \leq V \leq 1. \quad (10.4)$$

The partial transpose has one potentially negative eigenvalue, yielding

$$\mathcal{N}_B = \max \left[0, \frac{V(1 + 2|\gamma|) - 1}{4} \right]. \quad (10.5)$$

For the frozen H2 axes,

$$|S_{\text{frozen}}| = \sqrt{2}V(1 + \Re\gamma). \quad (10.6)$$

The optimized two-qubit CHSH diagnostic, from the two largest eigenvalues of $T^T T$, is [7]

$$S_{\text{max}} = 2V\sqrt{1 + |\gamma|^2}. \quad (10.7)$$

This optimized quantity diagnoses recoverable phase/correlation geometry; it does not replace the frozen primary witness.

For isotropic impurity $\gamma = 1$,

$$\mathcal{N}_B > 0 \iff V > \frac{1}{3}, \quad S_{\text{max}} > 2 \iff V > \frac{1}{\sqrt{2}}, \quad (10.8)$$

which is the familiar Werner-family separation between entanglement and CHSH violation [6, 7]. For pure dephasing $V = 1$, every $|\gamma| > 0$ is entangled, while the fixed real-axis witness requires

$$\Re\gamma > \sqrt{2} - 1 \quad (10.9)$$

for the frozen positive-sign orientation. Thus entanglement and violation of one frozen witness are distinct thresholds.

10.3 Bell-diagonal fidelity guardrails

For an arbitrary Bell-diagonal source, entanglement occurs iff the largest Bell population exceeds $1/2$. A singlet fidelity F alone guarantees the fixed canonical CHSH violation against worst-case Bell-diagonal contamination only if

$$F > \frac{1 + 1/\sqrt{2}}{2} \approx 0.8535533906. \quad (10.10)$$

Under the narrower depolarizing model the threshold is

$$F > \frac{1 + 3/\sqrt{2}}{4} \approx 0.7803300859. \quad (10.11)$$

The noise-model assumption changes the certificate.

10.4 Noisy GHZ family

Define

$$\rho_G(\gamma) = \frac{1}{2} \left(|000\rangle\langle 000| + |111\rangle\langle 111| + \gamma |000\rangle\langle 111| + \gamma^* |111\rangle\langle 000| \right) \quad (10.12)$$

and

$$\rho_G(V, \gamma) = V \rho_G(\gamma) + (1 - V) \frac{1}{8} \mathbb{I}_8. \quad (10.13)$$

Across every one-versus-two cut,

$$\mathcal{N}_{\text{cut}} = \max \left[0, \frac{V(1 + 4|\gamma|) - 1}{8} \right]. \quad (10.14)$$

For the frozen Mermin operator,

$$M = 4V\Re\gamma, \quad |M| = 4V|\Re\gamma|. \quad (10.15)$$

For isotropic noise,

$$\mathcal{N}_{\text{cut}} > 0 \iff V > \frac{1}{5}, \quad |M| > 2 \iff V > \frac{1}{2}. \quad (10.16)$$

For pure dephasing, the two-sided frozen Mermin violation requires $|\Re\gamma| > 1/2$. Noisy single- and pair-party marginals remain no-signaling, while formerly forbidden parity sectors fill continuously.

10.5 Swapping with impure sources

For two independent Werner-like inputs of visibilities V_{12} and V_{34} and an ideal middle Bell projection, the heralded outer sector has

$$V_{14}^{\text{out}} = V_{12}V_{34}. \quad (10.17)$$

Outer entanglement requires $V_{12}V_{34} > 1/3$, and outer CHSH violation requires $V_{12}V_{34} > 1/\sqrt{2}$. For equal input visibility,

$$V > \frac{1}{\sqrt{3}} \quad (10.18)$$

for outer entanglement, and

$$V > 2^{-1/4} \quad (10.19)$$

for outer CHSH violation.

10.6 Physical source impurity and no-signaling

Every positive normalized source state combined with local measurements remains no-signaling. An arbitrary source state need not have unbiased local restrictions. Therefore

$$\text{no-signaling} \Rightarrow \text{local marginals equal } \frac{1}{2}. \quad (10.20)$$

The 1/2 balance is a property of the selected source symmetry, not locality alone.

Physical two-qubit source states also satisfy the Tsirelson guardrail

$$S_{\max} \leq 2\sqrt{2} \quad (10.21)$$

[8]. Super-Tsirelson selected values remain evidence of selection bias or invalid normalization, not a noisy physical source kernel.

10.7 H10: explicit null outcomes and the detection loophole

Promote no-detection to a genuine local outcome:

$$\alpha, \beta \in \{-1, 0, +1\}. \quad (10.22)$$

With two settings on each wing, a deterministic local table specifies

$$A_0, A_1, B_0, B_1 \in \{-1, 0, +1\}, \quad (10.23)$$

so there are $3^4 = 81$ local extreme points.

For independent symmetric efficiency η , the full lossy frozen distribution is

$$q(\alpha, \beta | x, y) = \begin{cases} \eta^2 p(\alpha, \beta | x, y), & \alpha, \beta \neq 0, \\ \eta(1 - \eta)/2, & \text{exactly one outcome is null,} \\ (1 - \eta)^2, & \alpha = \beta = 0. \end{cases} \quad (10.24)$$

Deleting nulls cancels the common factor η^2 , so the coincidence-normalized CHSH score remains

$$|S_{\text{coincidence}}| = 2\sqrt{2} \quad (10.25)$$

for every $\eta > 0$. This selected score is not loophole free.

At $\eta = 0.8$, an explicit nonnegative mixture of 22 ternary local tables reproduces every raw probability cell while preserving the ideal coincidence subset. Thus the selected violation can coexist with a fully

Bell-local all-trial model.

10.8 Exact all-trial efficiency boundary

Use the Clauser–Horne event score [3]

$$J_{\text{CH}} = p(-, +|00) + p(-, +|01) + p(-, +|10) - p(-, +|11) \quad (10.26)$$

$$- p_A(-|0) - p_B(+|0). \quad (10.27)$$

Every Bell-local model satisfies $J_{\text{CH}} \leq 0$. For source visibility V and symmetric independent efficiency η ,

$$J_{\text{CH}}(\eta, V) = \frac{\eta^2}{2} (1 + \sqrt{2}V) - \eta. \quad (10.28)$$

Violation requires

$$\eta > \frac{2}{1 + \sqrt{2}V}. \quad (10.29)$$

At $V = 1$,

$$\eta_{\text{crit}} = \frac{2}{1 + \sqrt{2}} = 2(\sqrt{2} - 1) \approx 0.8284271247, \quad (10.30)$$

the Garg–Mermin boundary for this symmetric maximally entangled configuration [4]. A complete 81-vertex local-polytope LP confirms locality at the boundary and infeasibility above it within solver tolerance.

Assigning nulls the numerical value zero gives the valid but weaker all-trial score

$$|S_0| = 2\sqrt{2}V\eta^2, \quad (10.31)$$

so the ideal threshold is $\eta > 2^{-1/4}$.

10.9 Eberhard family and the 2/3 asymptote

Consider

$$|\psi_\epsilon\rangle = \cos \epsilon |00\rangle - \sin \epsilon |11\rangle. \quad (10.32)$$

Choose the positive event for setting zero as $|1\rangle$ and for setting one as

$$|v_\delta\rangle = \sin \delta |0\rangle + \cos \delta |1\rangle, \quad \tan^2 \delta = \tan \epsilon. \quad (10.33)$$

For the corresponding Clauser–Horne/Eberhard assignment [5], the relevant ideal probabilities are

$$p_{00} = \sin^2 \epsilon, \quad p_{01} = p_{10} = \frac{\sin^2 \epsilon}{1 + \tan \epsilon}, \quad p_{11} = 0, \quad (10.34)$$

while the two singles sum to $2 \sin^2 \epsilon$. Therefore

$$\eta_{\text{crit}}(\epsilon) = \frac{2(1 + \tan \epsilon)}{3 + \tan \epsilon}. \quad (10.35)$$

Hence

$$\lim_{\epsilon \rightarrow 0} \eta_{\text{crit}}(\epsilon) = \frac{2}{3}. \quad (10.36)$$

The useful event probabilities scale as $\sin^2 \epsilon$ and vanish in the same limit. The $2/3$ value is an ideal zero-background asymptote, not a universal finite-background laboratory guarantee.

10.10 GHZ raw-loss boundary

For independent local efficiencies,

$$M_{\text{raw}} = 4V \Re \gamma \eta_A \eta_B \eta_C, \quad |M_{\text{raw}}| = 4V |\Re \gamma| \eta_A \eta_B \eta_C. \quad (10.37)$$

All $3^6 = 729$ deterministic ternary three-party tables satisfy $|M| \leq 2$. For equal efficiencies and an ideal GHZ source,

$$\eta > 2^{-1/3} \approx 0.793700526. \quad (10.38)$$

This is the threshold of the frozen raw Mermin witness, not a global optimum over all multipartite inequalities.

10.11 Event-ready heralding firewall

A herald H may define the source ensemble only if

$$\mathbb{P}(H \mid \Lambda_{\text{src}}, a, b) = \mathbb{P}(H \mid \Lambda_{\text{src}}) \quad (10.39)$$

before the outer settings, or if equivalent independence is established by causal design and audit. Every heralded outer trial, including nulls, must remain in the data. A herald based on outer settings, outcomes, coincidence success, or desired correlations is postselection.

H9–H10 conclusion

Decoherence and impurity attenuate the frozen witnesses continuously and create broad entangled-but-nonviolating regions without signaling. Detection loss admits explicit Bell-local all-trial models below exact state- and witness-dependent thresholds. Coincidence-only violations cannot close the loophole; the primary statistic must retain nulls, singles, timing, and the full declared trial ensemble.

Final theorem layers, scope, and open debts

11.1 Layer I: ideal kernel theorem

Ideal SFT Bell-kernel theorem

Assume:

1. one local source event prepares the selected pure $L = 0, S = 0$ two-support channel;
2. the legal reduced source projector is Π_0 ;
3. each ideal local analyzer is a sharp rotationally covariant two-state detector with projectors $P_s(\mathbf{n})$;
4. eligible orthogonal support refinements contribute nonnegative additive relative intensity;
5. ideal completed record types are normalized, and the local branch-record maps act on disjoint spacelike factors.

Then

$$\mathbb{P}(s, t \mid \mathbf{a}, \mathbf{b}) = \frac{1}{4} (1 - st \mathbf{a} \cdot \mathbf{b}), \quad (11.1)$$

with unbiased ideal local marginals, perfect same-axis anticorrelation, operational no-signaling, and no preferred spacelike measurement order.

Measurement independence is not required to derive this conditional ideal distribution. It enters the next logical layer.

11.2 Layer II: Bell-local exclusion theorem

Bell-local exclusion theorem

Add the premise

$$P(a, b \mid \Lambda_{\text{src}}) = P(a, b). \quad (11.2)$$

Then the ideal distribution has no nonnegative normalized Bell-local factorization. For the predeclared CHSH axes it reaches

$$|S| = 2\sqrt{2} > 2. \quad (11.3)$$

The result preserves operational no-signaling while rejecting Bell-local factorizability under measurement independence.

11.3 Layer III: laboratory inference contract

To infer Bell nonlocality from a real lossy experiment, add the empirical protocol conditions:

1. the trial ensemble is declared independently of outer outcomes;
2. settings satisfy the frozen measurement-independence audit contract;
3. all local outcomes, including null, invalid, and saturation states, are retained;
4. singles, time tags, pairing rules, backgrounds, source impurity, filters, and stopping rules are predeclared and included in the primary analysis;
5. the primary statistic is all-trial and valid for the actual detector/source model.

Only at this layer can a laboratory data set be compared with the appropriate local polytope or inequality. Coincidence-normalized CHSH alone is insufficient.

11.4 What “causal locality but not Bell locality” means

Causal locality retained:

- source formation and detector interactions are local events;
- particle and apparatus support propagates causally;
- local settings enter only local detector maps;
- raw local marginals are independent of remote settings;

- no physical remote state update, signal, or preferred foliation is required;
- complete records are compared in the common future.

Bell locality rejected under the premise:

- the frozen joint law cannot be decomposed into a measurement-independent nonnegative mixture of locally factored response tables;
- no one global counterfactual answer table explains the statistics across all settings;
- the source relation constrains actual joint record sectors contextually rather than assigning independent local outcomes in advance.

The hostile program does not identify one uniquely metaphysical “violated assumption.” It proves the full Bell-local factorization fails under the frozen premises. Measurement dependence, remote-setting response, negative probability, and hidden postselection are separately constructed or quantified as escape routes and excluded from the frozen model.

11.5 The two ontology upgrades

11.5.1 H7: later relational joins

A universal common-birth-vertex explanation is too narrow. A later local joint interaction can join two previously independent source ancestries and create a heralded relational ledger sector. The object is tagged by a real middle record and is usable only with the corresponding classical information.

11.5.2 H8: genuine N -party objects

A complete relational state description cannot always be reduced to pairwise reduced objects. A GHZ source has separable pairwise reductions while retaining irreducible global coherence. SFT must admit N -party support objects when the prepared channel requires them.

These upgrades add neither fields nor dimensions. They broaden the arity and causal origin of legal relational support.

11.6 What has been derived

The following are frozen mathematical results within their named premises:

- the reduced selected-source projector $\Pi_0 = \frac{1}{4}(\mathbb{I} - \sigma_A \cdot \sigma_B)$;

- the unique ideal local detector law $D(\mathbf{n}) = \mathbf{n} \cdot \boldsymbol{\sigma}$;
- the support and ideal completed-event law $\frac{1}{4}(1 - st \mathbf{a} \cdot \mathbf{b})$;
- local null-complete no-signaling under arbitrary normalized local response channels;
- order-independent spacelike record completion;
- Bell-local polytope infeasibility and $2\sqrt{2}$ CHSH saturation;
- exact measurement-dependence escape metrics;
- delayed-choice, swapping, and GHZ consistency;
- exact Bell/GHZ threshold surfaces for the named noisy families;
- the maximally entangled detection threshold $2/(1 + \sqrt{2})$ for the frozen symmetric loss model;
- the Eberhard $2/3$ zero-background asymptote;
- the final all-trial experiment-entry contract.

11.7 What remains open

11.7.1 Microscopic source dynamics

- branching into the selected pure $L = 0, S = 0$ channel;
- exact relative-frame holonomy generation;
- native GHZ source selector and U_{GHZ} ;
- real source impurity, multipair, timing, and spectral distributions.

11.7.2 Microscopic detector dynamics

- exact U_{det} and U_{middle} ;
- material amplification and objective record formation;
- latency, saturation, dark counts, mode mismatch, and detector memory;
- absolute completion scale κ .

11.7.3 Environmental dynamics

- the coupling that produces external-history overlap γ ;

- $\gamma(t)$, material rates, spectral densities, and non-Markovian memory;
- source-specific tomography beyond the named noisy families.

11.7.4 Laboratory adjudication

No experiment is declared loophole free by this theoretical construction. A real claim requires the frozen trial protocol, raw null-complete data, source and detector calibration, timing and pairing audit, setting-independence audit, background model, and valid finite-data statistical analysis.

11.8 Program freeze

The source-of-truth construction hash is

```
de3faf4c6e4f4de92d127b86e1d1a09f01138143228e0e6947e5934d58ac4b19
```

The C5/C6 H1-entry contract hash is

```
f124fbbf445fad0fe00e5a3e33cafad5bda4ea8506d8f2653dc362ba25bc6d64
```

The H1–H10 hostile-program hash is

```
b23b0a3660cab3fd4814ea32a8833ba39eb2fe8c976f2856df3a58547b6e1274
```

Final audited conclusion

The Bell derivation is closed as a conditional mathematical program. The ideal kernel, no-signaling structure, contextual completion, frame-order consistency, Bell-local impossibility, multipartite extensions, noisy-family thresholds, and detection-loophole boundaries are explicit. The remaining work is microscopic source/detector dynamics, independent external adjudication, public narrative, and experiment-facing protocol design—not silent strengthening of the frozen claims.

Notation and Rosetta stone

Native SFT object	Representation	Meaning
H_v	common source record	permanent shared ancestry
$\Omega_{AB}^{(0)}$	legal zero-frame intersection domain	jointly admissible future pair histories
Π_Ω	orthogonal projector onto $\Omega_{AB}^{(0)}$	native legal source restriction
Π_0	reduced singlet/zero-total-frame projector	selected two-state $L = 0, S = 0$ relation
F_A, F_B	local frame transports	retarded closure-frame propagation
$R_{AB} = F_A^{-1} F_B$	relative holonomy	source relational frame datum
$P_s(\mathbf{n})$	rank-one local projector	sharp local analyzer branch
Q_{st}	tensor-product projector	one actual-context joint sector
μ_{st}	$\text{Tr}(\Pi_0 Q_{st})$	squared legal support measure
I_{st}	$\kappa \mu_{st}$	relative completion intensity
H_A^s, H_B^t	local record writes	irreversible local outcomes
H_C	ledger join	common-future complete record
γ	external-history overlap	remaining relational coherence

Exact frozen CHSH probability table

Let $r = 1/\sqrt{2}$. For contexts 00, 01, 10, $E = -r$, while $E_{11} = +r$. The sign probabilities are

$$p_{st}^{xy} = \frac{1}{4}(1 + stE_{xy}). \quad (\text{B.1})$$

Context	p_{++}	p_{+-}	p_{-+}	p_{--}
00	$(1-r)/4$	$(1+r)/4$	$(1+r)/4$	$(1-r)/4$
01	$(1-r)/4$	$(1+r)/4$	$(1+r)/4$	$(1-r)/4$
10	$(1-r)/4$	$(1+r)/4$	$(1+r)/4$	$(1-r)/4$
11	$(1+r)/4$	$(1-r)/4$	$(1-r)/4$	$(1+r)/4$

Final experiment-entry contract

A future empirical Bell claim derived from this canon must freeze before scoring:

1. a trial clock or valid event-ready herald;
2. setting-generation method and measurement-independence audit;
3. the complete local outcome alphabet, including null, invalid, and saturation states;
4. setting-resolved singles and null rates;
5. complete local time tags;
6. pairing, coincidence, offset, and window rules;
7. source impurity, multipair, background, and dark-count model;
8. the primary all-trial statistic;
9. all filtering and exclusion rules;
10. the sample size or valid sequential stopping rule;
11. drift, memory, block, and negative-control audits;
12. release of data, code, calibration, and sufficient aggregate counts.

Coincidence-normalized CHSH may be reported as a diagnostic. It cannot be the sole loophole-free statistic.

Gate and hash register

Gate	Frozen result	SHA-256
M1–M9	ideal construction freeze	de3faf4c6e4f4de92d127b86e1d1a09f01138143228e0e6947e5934d58ac4b19
C5/C6	H1-entry contract	f124fbbf445fad0fe00e5a3e33cafad5bda4ea8506d8f2653dc362ba25bc6d64
H1	Bell-local factorization kill	e2d123ef101fe363d545c7048b06dfe4e0718814111387e812257c5941358bbe
H2	frozen CHSH witness	00f06c2f2ba076262eac89f3cab00af997a4e0951927035cf4183cbf08ca3626
H3	detector/postselection stress	4bed65c5a066fd204cf7727033ae47512b4e836b9e361f31ca72694f6f95ba4a
H4	measurement-dependence escape	1ec083a7245f593b15dc85a56018578a59581fa2b7dcf727bfbce91321ccd5f1
H5	relativistic frame order	cb1c782ef23631815eb83c4eb313ad91e674fb120bc6c9d2d19248dc89e3b6f0
H6	delayed choice	f107919497a67d65150087826cc664821c101a20062969d1476c969f197b4813
H7	entanglement swapping	b5951bd84f31a924cc6d2b45e1f55a393fd2fdffcbd36bd97b8695cbaa47f188
H8	GHZ/ N -party	e9d23da8d7abb8a559499b0b4438147e2b4b81fa95261a3bcedd0c935b5edf98
H9	decoherence/impurity	3f974a160f665559d30f725ed567f2ef7164b2c04268c70161718dcd964ade03
H10	final detection attack	4b831c27e914f082d0aecd7b6ff07fe3a60ac6fb39e0b7db9e195cec13b5f521
Program	H1–H10 hostile freeze	b23b0a3660cab3fd4814ea32a8833ba39eb2fe8c976f2856df3a58547b6e1274

Version 1.0 correction register

Item	v0.1 issue	v1.0 resolution
A1	A projector sandwich was named as a projector without a commutation proof.	The native pair object is the legal zero-frame intersection; the sandwich is identified only as a positive filter when projectors do not commute.
A2	The isolated single-photon wording could be read as excluding recoil-assisted physical pair production.	The claim is restricted to an isolated one-null-support source with no external conserved support.
A3	Record branch maps duplicated a subsystem identity through inconsistent notation.	Tensor order and local branch Kraus maps are written explicitly.
A4	Linearity invoked informal regularity.	Nonnegative additivity is shown to imply monotonicity and exact linearity by rational squeezing.
A5	Unsharp effects were adjacent to the sharp orthogonality theorem.	POVM weights are separated from the sharp orthogonal support sectors.
A6	The H3 marginal artifact was called a probability shift of size $\sqrt{2}$.	It is identified as a signed-expectation shift; the associated $A = +$ probability shift is $1/\sqrt{2}$.
A7	The H4 information minimum lacked an in-document proof sketch.	A convex symmetry and failure-orbit reduction is included.
A8	The H7 middle-projector tensor order was implicit.	An embedded operator $\hat{\Pi}_\mu^{23}$ is defined on the ordered four-particle space.
A9	The noisy GHZ formula omitted an absolute value in the two-sided Mermin boundary.	$M = 4V\Re\gamma$ and $ M = 4V \Re\gamma $ are distinguished.

Item	v0.1 issue	v1.0 resolution
A10	The final theorem bundled kernel derivation, Bell exclusion, and empirical inference.	The conclusion is split into three logically independent layers.

Compact verdict rollup

PF1–PF2

The old singlet route contained a conditional theorem, but the angular intensity was quarantined until independently rederived.

M1 A complete local source event with the required conserved support is necessary.

M2/M2B

Generic no-overwrite does not force a literal braid; the selected pure $L = 0, S = 0$ channel supplies the source restriction as a premise.

M3–M4

A nonfactorizable, rotationally invariant, locally neutral zero-total-frame pair object is earned conditionally.

M5 The ideal local analyzer is uniquely $D(\mathbf{n}) = \mathbf{n} \cdot \boldsymbol{\sigma}$.

M6 The joint support law $\mu_{st} = \frac{1}{4}(1 - st\mathbf{a} \cdot \mathbf{b})$ is derived.

M7–M8

Actual-context completion and nonnegative additivity convert support measure into the normalized ideal event law.

M9 Local record maps commute and require no preferred spacelike order.

C5/C6

Null-complete no-signaling is a theorem; measurement independence is an explicit premise and audit obligation.

H1–H4

Bell-local factorization is killed; the measurement-dependence escape is quantified.

H5–H6

Frame-order and delayed-choice descriptions are consistent; order-conditioned subsets are postselection.

H7–H8

Later local relational joins and genuine N -party source objects are required.

H9 Decoherence and impurity thresholds are exact for the named noisy families.

H10 The final detection loophole is quantified; all-trial analysis and explicit nulls are mandatory.

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